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Mid Term Examination, Autumn 2022

CSE 1121 Computer Programming I

Total marks: 30 Time: 90 minutes

[Answer all the following questions. Figures in the right-hand margin indicate full marks.]

1. a) Name and describe four basic data types in C.

2 CLO1

b) `int main()`

13 CLO1

```
{  
    int p = 100000;  
    int q = 100000;  
    int result = p * q;  
    printf("%d", result);  
    return 0;  
}
```

Explain what is wrong with the above code and show the correct way to do it.

c) `double a = 0.1;`

1.5 CLO1

`if (a * 3 == 0.3)`

```
{  
    printf("Equal\n");  
}  
else  
{  
    printf("Not Equal\n");  
}
```

What is the output of the above code segment? Explain why does this output come.

d) Consider the following code that takes an integer input F which represents temperature in the Fahrenheit scale. It converts the temperature to the Celsius scale.

1 CLO1

```
int F;  
double cel;  
scanf("%d", &F);  
cel = (5 / 9) * (F - 32);  
printf("%.2f\n", cel);
```

Is the above code segment showing the correct output? If not, what should be done here?

e) Given the Basic of an employee. Write i) flowchart ii) algorithm/code to compute an employee's Gross pay and Net pay using the formulas-

4 CLO1

Gross = Basic + House Rent + Medical Allowance

Net = Gross - Tax

Tax is subtracted from the Gross only if an employee earns more than TK. 10000. Otherwise, deduct no Tax. Tax rate is 15% of Gross pay. House Rent is 60% of Basic and Medical Allowance is Tk. 700.

OR

You are given the height and width of two rectangles H_1 , W_1 and H_2 , W_2 . Write i) flowchart ii) algorithm/code to find out which rectangle is larger.

To find out the largest one-use area of rectangle. If area of both rectangles remains same, use the height for finding out the largest one. If both the rectangles are identical, print SAME instead.

2. a) A C program contains the following declarations and initial statements:

```
int i = 30, j = 15, k;
float x = 3.5, y = -1.1, z;
```

Determine the value of each of the following assignment expressions. Use the values originally assigned to the variables for each expression. Show the calculations.

- i) $k = i \% j$ iii) $y += (j/2)$
 ii) $z = k = x$ iv) $k = (j == 15) ? --i : j++$

- b) What would be the output of the following code segment: (use separate boxes for each digit, blank space and other symbols)?

```
int m = 8123;
float x = 34.567234600;
char str[30] = "Quality";
```

- i) `printf("%08d\n", m);`
 ii) `printf("%-10.2f\n", x);`
 iii) `printf("%7.3s\n", str);`

- c) A C program contains the following declarations and initial assignments:

```
int i = 13, j = 12;
double x = 7.8, y = -2.3;
char c = 'A', s[10] = "Morality";
```

Determine the value of each of the following expressions, which involve the use of library functions.

- i) `abs(i - 2 * j)` iv) `pow(2, 4)`
 ii) `ceil(x + y)` v) `islower(c)`
 iii) `floor(x - y)` vi) `strlen(s)`

- d) A student will not be allowed to sit in the exam if his/her attendance is less than 70%.

Take two integer input:

- i) Number of classes held,
 ii) Number of classes he/she attended.

Print the percentage of class attended and whether the student is allowed to sit in the exam or not.

Sample Input	Sample output
100 80	80.00%, Allowed
50 30	60.00%, Not Allowed

OR

You are given coordinates of two points in a 2D space. Determine If they are from the same quadrant.

Sample Input	Sample output
1 1 3 9	Yes
2 3 -3 -5	No

Note that point (2,3) lies in the 1st quadrant whereas point (-3, -5) belongs to the 3rd quadrant.


```

1. a) x = 1;
      y = 2;
      if (n > 5)
          if (n == 10)
              x += 2;
      else
          y -= 10;
      printf("%d %d ", x, y);

```

What will be the values of x and y if n assumes a value of i) 8 and ii) 3. Explain with rough calculations.

- b) Write a switch statement that will examine the value of a char type variable color and print one of the following messages depending on the value assigned to color.

Red, if color has a value 'R' or 'r'

Green, if color has a value 'G' or 'g'

Blue, if color has a value 'B' or 'b'

Not a Prime Color, if color has any other value

- c) Write C code segment to print all the even numbers from 1 to 100 inclusive in decreasing order using for loop. Rewrite the same using while loop.

- d) You are given a number X. Print all the odd divisors of X. A number N is a divisor of X if N divides X i.e. if we divide X by N the remainder is zero.

Sample Input	Sample output
16	1
21	1 3 7
28	1 7

Here in the first example divisors of 16 are: 1, 2, 4, 8 and 16. Here only the value 1 is odd.

OR

The term *evil number* is used to denote nonnegative integers that have an even number of 1s in their binary expansions. For example, 3 is an evil number since its binary expansion 11 has two 1s. The first few evil numbers are 0, 3, 5, 6, 9, 10, 12, 15, 17, 18, 20 ... Numbers that are not evil are then known as *odious numbers*. Write a C program that reads a positive integer N and then determine whether the given positive integer N is an *evil* number or *odious* number.

Input: Enter a positive integer: 3

Output: 3 is an Evil number.

Input: Enter a positive integer: 4

Output: 4 is an Odious number.

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Department of Computer Science & Engineering
Mid Term Examination, Autumn 2018
CSE 1121 Computer Programming I
Total marks: 30 Time: 90 Minutes

[Answer any **three** from the following questions. Figures in the right-hand margin indicate full marks.]

1. a) Name and describe four basic *data types* in C. 2
 - b) Determine which of the following are valid identifiers. If invalid, explain why? 2
 - i) CSE-1121 ii) mark\$ iii) inT iv) Computer programming
 - v) float vi) _abc vii) Long viii) 2mr
 - c) Determine which of the following are valid *character constants*. If invalid, explain why? 2
 - i) a ii) '\$' iii) '\n' iv) '\\'
 - v) 'T' vi) '\0' vii) 'xyz' viii) '\052'
 - d) Given three numbers A, B and C. Determine i) *algorithm* ii) *flowchart* to find the largest of these three numbers. 4
-
2. a) What is the difference between ++i and i++? Explain with example. 2
 - b) A C program contains the following declarations and initial assignments: 3

```
int i = 9, j = 6;
double x = 5.4, y = -1.5;
char c = 'A', s[10] = "IIUC CSE";
```

Determine the value of each of the following expressions, which involve the use of library functions.
 - i) abs(i - 2 * j) iv) pow(2, i)
 - ii) ceil(x + y) v) islower(c)
 - iii) floor(x - y) vi) strlen(s)
 - c) What are the differences between gets(str) and scanf("%s", str) where str is a character array? Explain with example. 2
 - d) Write a C program that input the time in seconds from keyboard and display the time in days (d), hours (h), minutes (m) and seconds (s) format as "ddd:hh:mm:ss" 3
Input: Enter the time in seconds: 240000
Output: Time in "ddd:hh:mm:ss" is 2:18:40:0
-
3. a) The following is a segment of a program: 1.5

```
y = 1;
if ( x >= 0 )
    if ( x > 0 )
        y = x + 1;
    else
        y = x - 1;
```

What will be the value of y if x equals to i) -2 ii) 0 and iii) 2?
 - b) Write a switch statement that will examine the value of an integer variable called *department* and print one of the following messages, depending on the value assigned to *department*: 1.5

```
CSE, if department have a value 1
EEE, if department have a value 2
ETE, if department have a value 3
Out of range of Engineering Faculty, if department has any other value
```

c) Write a C program segment to determine whether a given character input is vowel or consonant using i) *if..else* and ii) *switch* statements. 4

d) Write a C program that reads a year from the keyboard and determine whether it is leap year or not. [A year that is divisible by 4 and not divisible by 100 with the exception that years that are divisible by 400 are also leap year.] 3

Input: Enter the year: 2000

Output: 2000 is a leap year.

Input: Enter the year: 1900

Output: 1900 is not a leap year.

4. a) Determine how many times the body of each loop will be executed? 3

i)	ii)	iii)
int x=20,y=70;	int a = 6;	int x=3,y=9;
while(x<=y)	do	for(;x<=y;x++,y--)
{	{	{
y=y/x;	a = a - 6;	printf("%d\n",x);
}	} while(a>1);	}

b) Write C code segment to print all the *odd* numbers from 1 to 100 inclusive in decreasing order using *any* loop. 2

c) Suppose *a* and *b* are two integers. How can you swap (interchange) the value of these two variables in C? 2

d) Write a C program to calculate the sum of digit of a given positive integer number. 3
[For example, if we take 786 as input then the output should be 21 because 7+8+6=21]

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Mid Term Examination, Spring 2018
CSE 1121 Computer Programming I
 Total marks: 30 Time: 90 Minutes

Answer any **three** from the following questions. Figures in the right-hand margin indicate full marks.]

- 1) Draw the *block diagram* of a computer system. 2
- 2) Write short notes on: 2
 - i) Identifier ii) Symbolic Constant
- 3) Name and describe four *constants* in C. 2
- 4) Given working hours (Hours) and rate per hour (Rate) of an employee. Write i) *algorithm* 4
 ii) *flowchart* to compute an employee's Gross pay and Net pay using the formulas-

$$\text{Gross} = \text{Hours} * \text{Rate}$$

$$\text{Net} = \text{Gross} - \text{Tax}$$

Tax is subtracted from the Gross only if an employee earns more than TK. 10000/-.

Otherwise deduct no Tax. Tax rate is 15% of Gross pay.
- 5) What is a *unary operator*? Describe any three unary operators of C with example. 3
- 6) A C program contains the following declarations and initial statements: 2

```
int i = 12, j = 2, k;
float x = 3.5, y = 1.1, z;
```

Determine the value of each of the following assignment expressions. Use the values originally assigned to the variables for each expression:

 - (i) $z = k = y + 5$ (iii) $i += (j - 2)$
 - (ii) $k = i \% j$ (iv) $k = (j == 9) ? j++ : --i$
- 7) What will be the output of the following program: 2

```
#include <stdio.h>
int main ( )
{
    int i = 378;
    float x = 123.9453600;
    printf("%6d %2.4f \n", i, x);
    printf (" %-6d %12.4f \n", i, x);
}
```
- 8) Area of a triangle is given by the formula $\text{Area} = \sqrt{S(S-A)(S-B)(S-C)}$ where A, B and C are the sides of triangle and $2S = A + B + C$. Write a program to compute the area of the triangle given the values of A, B and C. 3
- 9) The following is a segment of a program: 2

```
x = 0; y = 0;
if (n > 5 )
    x = x + 2;
    y = y - 3;
printf(" %d %d ", x, y);
```

What will be the values of **x** and **y** if n assumes a value of i) 5 and ii) 10?

- b) Compare with example 3
- i) *switch* statement with the *nested if .. else* statement.
 - ii) *while* statement with the *do .. while* statement
- c) Write a switch statement that will examine the value of a character variable called *alphabet* and print one of the following messages, depending on the value assigned to *alphabet*: 2
- First Five Alphabet**, if *alphabet* has a value 'a', 'b', 'c', 'd' or 'e'
- Last Five Alphabet**, if *alphabet* has a value 'v', 'w', 'x', 'y' or 'z'
- Middle Alphabet**, if *alphabet* has any other value
- d) Suppose, Karnaphuli gas distribution company charges (*Energy charge*) the following rates to domestic users to discourage large consumption of gas: 3

Consumption Units	Rate of charges
0-150	Tk. 4 per unit
151-300	Tk. 6 per unit
301-499	Tk. 8 per unit
500 and above	Tk. 10 per unit

Now, Write a C program to read the number of units of gas consumed and print the amount to be paid by a customer.

- 4.a) Find out the *error* (if any) of the following program: 2

```
#include<studio.h>
Int main()
{
    int a, b, c
    scanf("%d%d",a,b);
    C= a+b;
    print("%d, "c);
    return 0
```

- b) A C program contains the following statements: 2

```
# include <stdio.h>
char a, b, c;
```

- i) Write appropriate *getchar* statements that will allow values for a, b and c to be entered into the computer.
- ii) Write appropriate *putchar* statements that will allow the current values of a, b and c to be written out of the computer (i.e, to be displayed).
- iii) Solve the problems (i), (ii) using a single *scanf* function and a single *printf* function rather than the *getchar* and *putchar* statements.

- c) Determine how many times the body of each loop will be executed. 3

<pre>i) x = 5; y = 50; while (x <= y) { y = y / x; }</pre>	<pre>ii) m = 1; do { m = m + 2; }while (m < 10);</pre>	<pre>iii) int i; for(i=0;i<=5; i=i+2) { printf("%d",i); }</pre>
--	---	--

- d) Write a C program to calculate the sum of digit of a given positive integer number. [For example, if we take 786 as input then the output should be 21 because 7+8+6=21] 3

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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MID TERM EXAMINATION, AUTUMN-2017 SEMESTER: 1ST

COURSE CODE: CSE-1121 COURSE TITLE: COMPUTER PROGRAMMING-1

TIME: 1 HOUR AND 30 MINUTES

FULL MARKS: 30

[Answer any three questions from the following questions. Figures in the right-hand margin indicate full marks.]

GROUP -A

1. a) What is algorithm? Write the rules for constructing an algorithm. 2
- b) Write a C program to read a length in *centimeter* scale and convert it in *inch* scale. 4
Draw the *flow chart* of it.
- c) Name and describe four basic constant in C. 2
- d) Write short notes on: 2

i) Identifier ii) Data type

2. a) A c program contains the following declarations and initial statements: 2

```
int i=12, j=2, k;  
float x=3.5, y= 1.1, z;
```

Determine the value of each of the following assignment expressions. Use the values origin assigned to the variables for each expression:

- | | |
|------------|---------------------------|
| (i) k=x=y | (iii) i += (j-2) |
| ii) z= i%j | (iv) k = (j == 5) ? i : j |

- b) Write the output of the following program: 2.5

```
#include<stdio.h>  
main() {  
float f=521.454333;  
char arr[20]="IUC CSE";  
int b=9876;  
printf("1#\n");printf("%06d\n", b);  
printf("2#\n");printf("%012.4f\n",f);  
printf("3#\n");printf("%*.*f\n",12,3,f);  
printf("4#\n");printf("%s\n",arr+5);  
printf("5#\n");printf("%#X\n",b); }
```

- c) What is the difference between Z++ and ++Z? Explain with example. 2
- d) Write a C program to evaluate the polynomial: 3.5

$$s = \left(\frac{x^2 + 1}{x}\right) + \left(\frac{x^2 + 1}{x}\right)^2 / 2 + \left(\frac{x^2 + 1}{x}\right)^3 / 3 + \left(\frac{x^2 + 1}{x}\right)^4 / 4 + \left(\frac{x^2 + 1}{x}\right)^5 / 5$$

Where the value of x is given from the keyboard.

3. a) A C program contains the following declarations and initial statements: 2

```
int i=4, j=16;  
double x=6.4, y=-1.1;  
char c='h', s[10]="Programming is Fun!!";
```

Determine the value of each of the following expressions, which involve the use of library function:

- | | |
|-----------------|----------------|
| i. islower (c) | iii. pow (i,3) |
| ii. floor (x-y) | iv. strlen (s) |

5/4

4345

- b) Which printf() will be executed in the following code segment:

```
int p, q;
p = 12;
q = -3;
if(p>0)
    if(q>0)
        printf("hello");
    else
        printf("world");
else
    printf("dhaka");
```

- c) Find out the error (if any) of the following program:

```
include<studio.h>
int Main()
{
    int b, goto;
    Char a;
    scanf("%d %c",&b,a)
    goto=b+5;
    Printf("%d",b);
    return 0
```

- d) Determine which of the followings are valid identifiers. If invalid, explain why?

i) 1AM ii) @fd zd iii) char
iv) JOB\$ v) @iiuc vi) "abcdef"

- e) Write a C program to input electricity unit charges and calculate total electricity bill according to the given condition:

- For first 50 units Tk. 0.50/unit
- For next 100 units Tk. 0.75/unit
- For unit above 150 Tk. 1.50/unit

An additional surcharge of 20% is added to the bill.

4. a) Determine how many times the body of each following loop will be executed:

i)	ii)	iii)
int x=13,y=21;	int x=10,y=90;	int a=23;
for(;x<=y;x++,y--)	while(x<=y)	do
{	{	{
printf("%d\n",x);	y=y/x;	a=a-9;
}	}	} while(m>1);

- b) Write a C program to calculate the sum of digit of any integer number. [For example, if we take 123 as input, then the output should be 6 because 1+2+3=6]

- c) White the purpose of the following statements with example:

i) goto ii) continue iii) break

- d) Write a switch statement that will examine the value of an integer variable called bus and print one of the following messages, depending on the value assigned to bus:

Agrabad, if bus have a value 1
Chawkbazar, if bus have a value 2
Boddharhat, if bus have a value 3
Out of range of transport facility, if bus has any other value

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Department of Computer Science & Engineering
Mid Term Examination, Autumn 2017
CSE 1201 Structured Programming
Total marks: 30 Time: 90 Minutes

Answer any *three* from the following questions. Figures in the right-hand margin indicate full marks.]

Part A

Write down the basic *structure* of a C program.

2

Write short notes on:

2

i) Identifier ii) Symbolic Constant

Name and describe four basic *data types* in C.

2

Given three numbers A, B and C. Write i) *algorithm* ii) *flowchart* to find the *largest* of these three numbers.

4

What is a *unary operator*? Describe any three unary operators of C with example.

2

A C program contains the following declarations and initial statements:

2

`int i = 12, j = 5, k;`

`float x = 3.5, y = -1.1, z;`

Determine the value of each of the following assignment expressions. Use the values originally assigned to the variables for each expression:

i) `z = i % j`

iii) `y += (j/2)`

ii) `z = k = x`

iv) `k = (j == 5) ? i++ : --j`

What do you mean by *precedence* and *associativity* of operators? Explain with example.

2

Write a program to compute the real roots of a quadratic equation $ax^2 + bx + c = 0$. The roots are given by the equations, $x_1 = -b + \frac{\sqrt{b^2 - 4ac}}{2a}$ and $x_2 = -b - \frac{\sqrt{b^2 - 4ac}}{2a}$. The program should request for the values of the constants a, b and c and print the values of x_1 and x_2 . Use the following rules:

4

i) No solution, if a is zero

ii) There are no real roots, if $b^2 - 4ac$ is negative

iii) There is only one root, if $b^2 - 4ac = 0$

iv) Otherwise, there are two real roots x_1 and x_2

Which `printf()` will be executed in the following code segment:

1

```
int p, q;  
p = 12;  
q = -3;  
if(p > 0)  
    if(q > 0)  
        printf("hello");  
    else  
        printf("world");  
else  
    printf("HUC");
```

- b) Write a *switch* statement that will examine the value of an integer variable called *bus* and print, of the following messages, depending on the value assigned to *bus*:
- Agrabad, if *bus* have a value 1
 Chawkbazar, if *bus* have a value 2
 Boddharhat, if *bus* have a value 3
 Out of range of transport facility, if *bus* has any other value.
- c) Compare with example
- i) *if.. else* statement with the *?:* operator.
 - ii) *switch* statement with the *nested if.. else* statement
- d) Write a program which receives the *marks* of a student and display the grade according to the following classification:

<u>Marks</u>	<u>Grade</u>
80 to 100	Honors
60 to 79	First Division
45 to 59	Second division
30 to 44	Third Division
0 to 29	Fail

- 4.a) Find out the *error* (if any) of the following program:

```
include<studio.h>
int Main()
{
  int b, c;
  Char a;
  scanf("%d %c", &b, a)
  C=b+5;
  Printf("%d", b);
  return 0
}
```

- b) Determine which of the following are valid identifier. If invalid, explain why?

i) 2nd ii) fd zd iii) _char
 iv) Int v) @iiuc vi) "abcdef"

- c) A C program contains the following declarations and initial statements:

```
int i=4, j=16;
double x=6.4, y=-1.1;
char c='h', s[10]="Programming is Fun!!";
```

Determine the value of each of the following expressions, which involve the use of library function:

i. islower (c) iii. pow (i,3)
 ii. floor (x-y) iv. strlen (s)

- d) An electricity board charges (*Energy charge*) the following rates to domestic users to discourage large consumption of energy:

For the first 75 units: Tk. 4.0 per unit
 For Next 325 units: Tk. 5.7 per unit
 Beyond 400 units : Tk. 9.3 per unit

All users are charged a minimum of Tk. 100/- for Energy charge. If the total cost for Energy charge is more than Tk. 8000/- then an additional surcharge of 15% is added. In addition to this Energy charge, all users have to pay Tk. 15/- as *Demand charge* and Tk. 10/- as *Service charge*. So, **Net Bill = Energy charge + Demand charge + Service charge**
 Write a C program to read the number of units consumed and print out the *Net Bill*.

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INTERNATIONAL ISLAMIC UNIVERSITY CHITTAGONG
 Department of Computer Science and Engineering (CSE)
 Mid-Term Examination, Spring, 2017
 Course Code: CSE-1201 Course Title: Structured Programming
 Full Marks: 30 Time: 1 Hour 30 minutes

[Answer any three of the following questions (Figures at right margin illustrate mark)]

3-10-30

1. a) What is the basic difference between structured and unstructured programming? C is a structured programming language. Justify your answer. 2
- b) What are the major components of C program? What significance is attached to use `int` instead of `void` in main function of a C program? 2
- c) What is meant by C token? Identify the C tokens of following C program.

```
#include <stdio.h>
int main() {
    int a=10, b=5;
    printf("a+b = %d\n", a+b);
    return 0;
}
```

2
- d) Determine which of the following are valid identifiers. If invalid, explain why? 2
 i) `Float` ii) `_CSE` iii) `return0` iv) `salary_12` v) `mark`
 vi) `double` vii) `CSE 1201` viii) `Spring#2017` ix) `auto`
- e) Determine which of the following are valid constants. If a constant is valid, specify whether it is integer (specify base) or real. For invalid constants, explain why? 2
 i) `33,545` ii) `0277` iii) `23/560L` iv) `0xA5C` v) `3.3E+2.4`
 vi) `0Xae5` vii) `278C01L` viii) `2.3L`
2. a) What is expression? A C program contains the following declarations and initial assignments: 1

```
int i = 15, j = 15, k;
```

```
float x = 11.5, y = -1.2, z;
```

 Determine the value of each of the following assignment expression. Use the values originally assigned to the variables for each expression.
 i) `k = (x - y)` iv) `k = 2 - i / j`
 ii) `z = i % j` v) `k = (j -- 5) ? i-- : j++`
 iii) `k = x-1` vi) `i += (j > 5) ? j : i`
- b) Write the output of the following C program: 1.5

```
#include <stdio.h>
int main() {
    int i=2, j=4, k=5;
    if(k%2 == 1) {
        i=j;
        printf("i = %d\n", i);
        j=k;
    }
    else if(k%2 == 0) {
        printf("j = %d\n", j);
        printf("k = %d\n", --k);
    }
    return 0;
}
```

3A

3A

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- c) Write the exact output of the following C program:

```
#include<stdio.h>
int main() {
    char s[] = "Printing 2017";
    float f = 113.57040;
    int i = 456;
    printf(":%03d:%-05d:\n", i, i);
    printf("%. *s\n", 2, s);
    printf("%o\n%x", i, i);
    printf(":%*.*f\n", 4, 3, f);
    printf("%g\n%.4e", f, f);
    return 0;
}
```

- d) What is meant by the operator precedence and associativity?

- e) What is the basic difference between `gets(X)` and `scanf("%s", X)` statements. Where X is character string.

- 3 a) An electricity board charges the following rates to domestic users to discourage the large consumption of energy:

For the first 50 units : Tk. 3.70 per unit
 For next 20 units : Tk. 3.00 per unit
 Beyond 70 units : Tk. 4.00 per unit

All users are charged a minimum of Tk. 150. If the total cost is more than Tk. 100 then an additional surcharge of 10% will be added. Write a program to read the number of units consumed and print out the total charges in BDT.

- b) Write the output of the following C program:

```
#include<stdio.h>
int main() {
    char id='F';
    switch(id) {
        case 'C': printf("CSE\n");
        case 'F': printf("FSE\n");
        case 'I': printf("IUC\n");
                break;
        default : printf("Error\n");
    }
    return 0;
}
```

- c) Find the output for the following program:

```
#include<stdio.h>
int main() {
    signed char c=132;
    signed char ch=-133;
    printf("c = %d\nch = %d", c, ch);
    return 0;
}
```

- d) Write a `switch` statement that will examine the value of an integer variable called `indicator` and print one of the following messages, depending on the value assigned to `indicator`.

CSE, if `indicator` have a value of 0 or 1
 FEE, if `indicator` have a value of 2
 LEE, if `indicator` have a value of 3 or 4
 PEM, if `indicator` have a value of 5 or 6
 or or or, if `indicator` has any other value.

Given following information:

Attendance	Marks
90% and above	10
85% to less than 90%	9
80% to less than 85%	8
75% to less than 80%	7
70% to less than 75%	6
65% to less than 70%	5
60% to less than 65%	4
Less than 60%	3

Sample input and output:

Enter attended days: 20
Attendance Percentage: 47.6
Marks: 6

Write a program to calculate the marks and attendance percentage of a student for a three contract hour's course in a semester. The maximum no. of classes for this course is 42. The attended days of a student accepts as a input positive integer number and you have to print the percentage and marks for given attendances.

b) What are unary operators? Write purposes of the following c programming statements:
i) `sizeof(int);` ii) `a++;` iii) `--a;`

c) Suppose, FSE decided to host the "IIUC Tech Festival" in 29th February of each year, but this day does not come every year. It comes only those years which are perfectly "divisible by 4 and which are not divisible by 100" or "divisible by 400". Now, you have to write a program to find out the given year is "IIUC Tech Festival year" or "Ordinary year".

Sample input:	Sample output:
2017	Ordinary year
2018	IIUC Tech Festival year
2100	Ordinary year

d) What do you mean by type casting? Elucidate the different type casting in C programming with example.

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International Islamic University Chittagong
 Department of Computer Science & Engineering
Mid Term Examination, Autumn 2016
CSE 1201 Structured Programming
Total marks: 30 Time: 90 Minutes

[Answer any *three* from the following questions. Figures in the right-hand margin indicate full marks.]

1. a) Who have developed C and where? 1
 b) Write short notes on: 2
 i) Escape Sequence ii) Symbolic constant 3
 c) Name and describe the four basic types of *constants* in C. 4
 d) Given three numbers A, B and C. Determine i) *algorithm* ii) *flowchart* to find the largest of these three numbers. 4

2. a) What is the difference between ++i and i++? Explain with example. 2
 b) A C program contains the following declarations and initial assignments: 3
 int i = 11, j = 9;
 double x = 4.5, y = -5.1;
 char c = 'B', s[10] = "IUC CSE";
 Determine the value of each of the following expressions, which involve the use of library functions.
 i) abs(i - 2 * j) iv) pow(2, i)
 ii) ceil(x + y) v) islower(c)
 iii) floor(x - y) vi) strlen(s)
 c) What are the differences between gets(str) and scanf("%s", str) where str is a character array? Explain with example. 1.5
 d) Write a C program to calculate the roots of the quadratic equation $ax^2 + bx + c = 0$ where a, b and c are read as input. 3.5

3. a) The following is a segment of a program: 2
 x = 10;
 y = 20;
 if (n > 0)
 x = x - 5; y = y + 7;
 printf("x = %d y = %d", x, y);
 What will be the values of x and y if n assumes a value of i) 2 and ii) 0 ?
 b) Write a switch statement that will examine the value of an integer variable called department and print one of the following messages, depending on the value assigned to department: 2
 CSE, if department have a value 1
 EEE, if department have a value 2
 ETE, if department have a value 3
 Out of range of Engineering Faculty, if department has any other value
 c) Compare the use of the switch statement with the use of nested if..else statements with examples. Which one is more convenient? 2.5
 d) Write a C program that reads a year from the keyboard and determine whether it is leap year or not. [A year that is divisible by 4 and not divisible by 100 with the exception that years that are divisible by 400 are also leap year.] 3.5

4.a) Determine which of the following are valid identifiers. If invalid, explain why?

- i) 51a ii) xy z iii) _XYZ iv) Float
v) Structured_Programming vi) "CSE"

b) What will be the output of the following program:

```
#include<stdio.h>
int main()
{
    int x = 18745;
    float z = 78.12542;
    char s[20] = "IIUC CSE 1201";

    printf("%10.2f\n",z);
    printf("%08d\n",x);
    printf("%+8d\n",x);
    printf("%-12.2e\n",z);
    printf("%20.10s\n",s);

    return 0;
}
```

c) Suppose **a** and **b** are two integers. How can you swap (interchange) the value of these two variables in C?

d) Write a C program that computes an employee's Gross pay and Net pay using the formulas-

Gross = Basic + House rent + Medical allowance

Net = Gross - Tax

Tax is subtracted from the Gross only if an employee earns more than TK. 10000. Otherwise deduct no Tax. Tax rate is 7% of Gross pay. House rent is 50% of Basic and Medical allowance is TK. 500.